

AIM:Implement Quick sort and Merge sort using Divide and conquer technique and observe the execution time for various input sizes (Average, Worst and Best cases)

Program : C program to implement Merge Sort Algorithm

```
#include<stdio.h>
#include<stdlib.h>
void mergesort(int arr[],int low,int high);
void merge(int arr[],int low,int mid,int high);
void mergesort(int arr[],int low,int high){
if(low>=high)
return;
int mid = (low+high)/2;
mergesort(arr,low,mid);
mergesort(arr,mid+1,high);
merge(arr,low,mid,high);
}
void merge(int arr[],int low,int mid,int high){
int size = high-low+1;
int *temp = (int*)malloc(size*sizeof(int));
int counter=0;
int left=low;
int right=mid+1;
while(left<=mid && right<=high){
if(arr[left]<=arr[right])
temp[counter++] = arr[left++];
else
temp[counter++] = arr[right++];
}
while(left<=mid)
temp[counter++] = arr[left++];
while(right<=high)
temp[counter++] = arr[right++];
for(int i=0;i<size;i++)
arr[i+low] = temp[i];
free(temp);
}
```

```
void main(){
int n;
scanf("%d",&n);
int arr[n];
for(int i=0;i<n;i++){
scanf("%d",&arr[i]);
}
printf("Array before sorting : ");
for(int i=0;i<n;i++)
printf("%d ",arr[i]);
printf("\n");
mergesort(arr,0,n-1);
printf("Array after Mergesort : ");
for(int i=0;i<n;i++)
printf("%d ",arr[i]);
}
```